



JAKUB RADLICKI

Technical Artist | Procedural Generation & Rendering

Portfolio: [View Portfolio](#) | Email: architecturedesigns.pl@gmail.com | LinkedIn: [jakub-radlicki](#) | Born: 2002.06.09 | Poland

Video Showcases: [Showcase 1](#) | [Showcase 2](#) | [Showcase 3](#)

PROFILE SUMMARY

I'm a 3D artist and tech enthusiast who loves bridging the gap between heavy math logic and great visuals. Most of my technical skills, like graphics programming and pipeline development, are completely self-taught alongside my architecture degree. I've built my own custom rendering engines using Vulkan and C++, set up massive procedural generation systems using Python and Geometry Nodes, and crushed 50-million polygon scans into lightweight real-time assets. I'm really passionate about pushing Unreal Engine 5 as far as it can go.

EXPERIENCE

Technical Artist / Systems Developer

FREELANCE (2020 - PRESENT) | ARCHITEKCI I DREWNO (2020.01 - PRESENT)

- Built a custom infinite ocean shader that handles overwater and underwater transitions seamlessly.
- Put together complex mathematical shaders using things like RVT blending, POM, and frustum-culling.
- Wrote my own standalone rendering engine from scratch using C++ and Vulkan.
- Took a massive 50-million polygon photogrammetry scan and baked it down to a 35k asset that runs perfectly in-browser.
- Scripted out procedural generation tools using Python and nodes to completely automate structural modeling tasks.

Contract 3D Artist & Technical Freelancer

J7 (2025.10 - 2026.02) | 3NGAGE (2024.07 - 2024.09) | KNWLY (2022.01 - 2024.07)

- Set up logic-driven batch rendering pipelines that automatically swap out materials and cameras for hundreds of meshes.
- Created custom shaders and optimized 3D assets for different studios while sticking strictly to their performance budgets.
- Helped external studios figure out why their asset pipelines were choking and fixed them using procedural tools.
- Handled high-end interior design layouts and modeled custom 3D furniture for a few commercial clients.

EDUCATION

Bachelor of Architecture

- Leveraged architectural data and geometry systems to enhance 3D pipelines, including implementing IFC data filtering to automate specific architectural parameter exports.

TECHNICAL SKILLS

Unreal Engine 5

Blender (Geometry Nodes)

C++ / Python

Vulkan / GLSL / HLSL

Procedural Content Generation (PCG)

Extreme Optimization

Shader Development

SOFTWARE & TOOLS

3D Modeling, Sculpting & Texturing: Blender, Nomad, Mudbox, Houdini, Autodesk Maya, Quixel Mixer.

Engines & Rendering: Unreal Engine 5 (UE5), Unity, Godot, Vulkan API, OpenGL, V-Ray, Cycles, Eevee.

CAD, Architecture & Parametric: AutoCAD, Vectorworks, Revit, Rhinoceros 3D, Grasshopper, SketchUp, Archicad, Robot, Rwin, Enscape, D5.

Development, Version Control & Automation: C++, Python, C#, HLSL, GLSL, SPIR-V, JavaScript, Git, GitHub, Docker, n8n, CMake, Visual Studio, VS Code.

Compositing, Video & Motion Graphics: DaVinci Resolve, Blender Compositor.

UI/UX, Web & 2D Design: Spline 3D, Affinity, Figma, Photoshop, Illustrator, Lightroom, CorelDRAW.